

```

#include<stdio.h>
#include<math.h>
#define maxsize 4
#define TRUE 1
#define FALSE 0

struct CQueue
{
int rear, front;
int item[maxsize];
};

struct CQueue cq;

void insert(int);
int delete();

int empty();

void main()
{
char ch;
int c,x,i;
cq.rear=maxsize-1;
cq.front=maxsize-1;
do
{
printf("MENU\n");
printf("1. INSERT \n 2. DELETE\n 3. Display\n ");
scanf("%d",&c);
switch(c)
{
case 1: printf("enter the no to be inserted\n");
scanf("%d",&x);
insert(x);
break;
case 2: x=delete();

```

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        if(x==-1) printf("Queue underflow\n");
        else
            printf("deleted element is %d",x);
        break;
case 3: printf("elements are\n");
        if(cq.front==maxsize-1)
        {
            for(i=0;i<=cq.rear;i++)
            {
                printf("%d ", cq.item[i]);
            }
        }
        else
        {
            for(i=cq.front+1;i<=cq.rear;i++)
            {
                printf("%d ", cq.item[i]);
            }
        }

        break;
    }
    printf("do you wish to continue\n");
    scanf(" %c",&ch);
}while(ch!='n');
}

```

```

void insert(int x)
{
    if(cq.front==((cq.rear+1)%maxsize))
    {
        printf("Queue overflow\n");
    }
    else
    {
        cq.rear=(cq.rear+1)%maxsize;
        cq.item[cq.rear]=x;
    }
}

```

```
}
```

```
int delete()
```

```
{
```

```
if(empty())
```

```
{
```

```
return(-1);
```

```
}
```

```
else
```

```
{
```

```
cq.front=(cq.front+1)%maxsize;
```

```
return(cq.item[cq.front]);
```

```
}
```

```
}
```

```
int empty()
```

```
{
```

```
if (cq.rear==cq.front)
```

```
return TRUE;
```

```
else
```

```
return FALSE;
```

```
}
```